

JANANI ASOKAN

[Github](#) | [Portfolio](#) | [Linkedin](#) | ☎ 6383495030 | ✉ jananiasokanse@gmail.com

SUMMARY:

Software Engineer with 2+ years of experience in API Automation, Backend Development, and Systems Engineering. Strong knowledge of operating systems, file systems, networking, and data structures. Hands-on experience in API testing, integration testing, system testing, regression testing, and automation frameworks. Experienced in cloud technologies, distributed systems, and scalable application development.

EXPERIENCE:

Cloud API Automation Engineer (Signify formerly known as Philips) Bengaluru, India July 2025 – Present

- Built and maintained **scalable backend API testing frameworks** using Python and Pytest for validating RESTful services across distributed environments (dev/staging/prod)
- Designed **automation pipelines**, improving test coverage and reducing manual validation effort by 65%.
- Validated real-time distributed data flow across MQTT brokers, cloud APIs, and mobile systems, ensuring system consistency and reliability for over 89% of connected devices or processes.
- Implemented robust authentication systems including token lifecycle and session management as well as fault-tolerant mechanisms, retry logic, and latency validation

Sidhar Institute of Technology Assistant Professor – AI/ML

Dec 2025 - Present

- Delivered training sessions in **Artificial Intelligence and Machine Learning**, covering core concepts and practical applications.
- Mentored students on real-world projects and guided them in building ML models.
- Assisted students in improving employability by optimizing resumes for **ATS (Applicant Tracking Systems)**.
- Provided hands-on guidance for platforms like **Naukri**, ensuring better visibility to recruiters.
- Supported students in placement preparation, resulting in improved shortlisting and interview success rates.

Frontend Developer (SkyThrinethra)

June 2024 – Feb 2025

- Engineered dynamic, reusable React.js components with efficient state management, enabling modular UI architecture that reduced development time for new features by improving code reusability across the platform.
- Integrated RESTful APIs built on Node.js and Express.js into frontend interfaces, achieving a 78% improvement in real-time data rendering speed, optimizing API call handling, and delivering a seamless user experience across all application modules.
- Leveraged MongoDB as the backend data layer, collaborating on schema design that supported complex frontend data flows while maintaining application scalability and performance.

MTech Computer Science and Engineering | **CGPA: 8.4** | Vellore Institute of Technology, Vellore

July 2021 – July 2026

XII KSR school, Madhanur

Percentage 93.4%

X KSR school, Madhanur

Percentage 92.5%

SKILLS:

Languages: Python, C++

AI Tech: Machine Learning, Deep Learning, Pandas, NumPy, Power BI, Matplotlib

Computer Science Fundamentals: Data Structures and Algorithms, Operating Systems, AWS, Computer Networks

Testing & Automation: MQTT, Postman, Pytest, Firmware Testing

Software: GitHub, VS Code, Google Colab, Wireshark, Packet sender

Soft skills: Highly Responsible, Problem-solving skills, Effective communication, Self-motivated

PROJECTS:

AI-Powered Healthcare Orchestration System | AI, ML, Kafka, OpenCV, Python, React, PostgreSQL [\[Live Demo\]](#)

- Engineered an innovative distributed AI platform leveraging Kafka and PostgreSQL to enable real-time healthcare coordination, driving enhanced efficiency and care delivery.
- Designed **event-driven pipelines** for processing and routing emergency data
- Developed **backend services and APIs** for communication and decision support and Improved response efficiency through predictive analytics and automation

IoT-Based Response Drone for Search and Rescue | DL, YOLO V8, CNN, Restful APIs [\[Github\]](#)

- Developed an AI-powered drone system using Arduino for real-time disaster response and search operations
- Integrated an AI-driven **image processing** module to identify stranded individuals in disaster-affected areas with 95% accuracy.
- Conducted simulation testing under flood scenarios, validating system reliability for remote deployment

WebDigestAI - Efficient Content Digesting with AI-driven Insights | NLP, Python, LangChain, LangGraph [\[Live Demo\]](#)

- Designed and implemented a retrieval-augmented generation (RAG) pipeline utilizing LangChain and transformer-based large language models for large-scale document processing.
- Designed **scalable ingestion and retrieval pipelines** handling 1000+ data sources
- Integrated LLM APIs with optimized prompting and context retrieval for high-accuracy responses. Focused on efficient data flow, latency reduction

AWARDS & CERTIFICATIONS:

- **PitchWar Hackathon finalist** secured 3rd place by developing an IoT-enabled drone system with AI-driven image processing - Best Innovator Award
- **ISRO Hackathon finalist** - ISRO Hackathon Finalist — Achieved 87.2% accuracy on 30,000 lunar images using YOLOv8
- **Postman API (Postman)**
- **Artificial Intelligence Accenture**
- **Research on Federated Learning for Autonomous Vehicle Detection**— Developed a decentralized AI model
- **ISTQB Certified Holder**